

Blackboard Pattern as a Tangential Block Field Viewed Computationally

Abstract

The Blackboard Pattern is commonly described as a software architecture in which independent knowledge sources contribute partial results to a shared workspace.

Within Tangential Block Geometry, this computational pattern is understood as the implementation-level appearance of a deeper geometric structure:

Blackboard Pattern = Tangential Block Field viewed computationally

The blackboard is not fundamentally a shared memory object, a database, or a globally materialized table. It is a bounded projective field in which otherwise independent relational blocks become jointly visible.

Each contributor retains its own complete identity and internal coordinate system. It projects only a bounded contribution into the shared field. A composition operator, represented by CONS, resolves how these contributions meet. Compatible contributions form a shared visible structure; conflicting or ambiguous incidences remain present as residual witnesses rather than being silently erased.

The computational blackboard is therefore the visible surface of tangency among autonomous block designs.

1. Conventional Blackboard Pattern

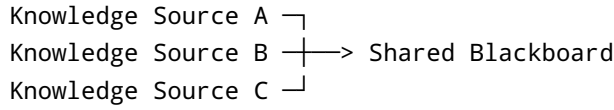
In a conventional Blackboard Pattern, several specialized modules operate independently.

Each module:

- observes some part of a problem;
- produces a partial result;
- posts that result into a shared workspace;
- may respond to results posted by other modules.

The blackboard allows multiple contributors to cooperate without requiring any one contributor to contain the entire model.

The general computational form is:



The shared board acts as a coordination surface.

This description is operationally useful, but it does not explain what the shared surface is geometrically, why independent contributions can coexist there, or how identity is preserved when many different structures project into the same bounded workspace.

Tangential Block Geometry supplies that missing foundation.

2. From knowledge sources to blocks

A knowledge source is interpreted geometrically as a bounded relational block or block design.

Let:

$$B_i \in \mathcal{B}$$

denote the relational structure contributed by source i .

A block may contain:

- points;
- relations;
- declarations;
- observations;
- rules;
- facts;
- operations;
- closure conditions;
- local coordinates;
- remote coordinates.

The block is autonomous.

It does not become part of the blackboard merely by existing.

It becomes visible only when projected into the shared field.

Define:

$$\Pi_{\mathbb{B}}(B_i) = S_i$$

where:

- $\mathbb{B}$ is the shared blackboard field;
- S_i is the visible contribution of block B_i .

The blackboard therefore contains projections of blocks, not necessarily the complete blocks themselves.

3. The blackboard as a field of visibility

The word **field** here means a shared bounded coordinate domain in which contributions may be located, compared, and composed.

It does not automatically mean an algebraic field with addition, multiplication, and inverses.

Define the Tangential Block Field as:

$$\mathbb{B} = (P_B, \mathcal{S}, \Gamma)$$

where:

- P_B is the set of visible blackboard positions;
- $\mathcal{S}$ is the family of projected contributions;
- Γ is the declared composition law.

Each contribution is a subset or structured occupancy of the field:

$$S_i \subseteq P_B$$

The visible blackboard state is:

$$B_{\text{visible}} = \Gamma(S_1, S_2, \dots, S_n)$$

Thus the blackboard is not one contributor and is not identical to the union of all underlying knowledge.

It is the current visible result of composing selected projections.

4. Full identity and visible position

A foundational distinction must be preserved:

Blackboard position is not complete identity.

In the earlier OMI blackboard model, CAR and CDR are full 32-bit ordered coordinates, while the 256-position OMI-Lisp plane is their bounded projection surface.

Let:

$$R_i$$

be a complete relational identity, and let:

$$p_i = \Pi_{\mathbb{B}}(R_i)$$

be its visible blackboard position.

It is possible that:

$$\Pi_{\mathbb{B}}(R_1) = \Pi_{\mathbb{B}}(R_2)$$

while:

$$R_1 \neq R_2 =$$

Several complete relations may project to one visible coordinate.

The projection is therefore many-to-one unless additional identity data is retained.

A complete identity may bind:

$$\text{Identity}(R) = (CAR, CDR, \text{resolver}, \text{orientation}, \text{active block}, \text{projection})$$

The blackboard coordinate answers:

Where is this relation currently visible?

It does not necessarily answer:

What is the complete relation?

The older blackboard specification makes this explicit: the eight-bit plane is a view, not the complete address.

5. CAR, CDR, and CONS

The computational field is organized around three roles:

CAR

CDR

CONS

CAR and CDR are complete opposing or complementary relational coordinates.

They may represent:

```
CAR:
local
source
declaration
observer
input
object

CDR:
remote
continuation
definition
other observer
output
relation
```

The exact interpretation depends on the active projection.

CONS resolves the relation between them.

The complete operation is:

$$CONS(CAR, CDR) = (R_{CAR,CDR}, \Pi_{\mathbb{B}}(R_{CAR,CDR}))$$

where:

- $R_{CAR,CDR}$ is the complete ordered relation;
- $\Pi_{\mathbb{B}}(R_{CAR,CDR})$ is its visible field coordinate.

The earlier computational model already describes CONS as performing both full CAR/CDR resolution and projection into the shared byte plane.

CONS is therefore not only a pair constructor.

It is:

- a tangent operation;
- a relation resolver;
- a projection operator;
- a boundary between complete identity and visible coordination.

6. CONS as tangency

In Tangential Block Geometry, tangency describes contact without loss of distinction.

Two blocks:

$$B_{CAR}$$

and:

$$B_{CDR}$$

may meet through:

$$\Lambda(B_{CAR}, B_{CDR})$$

without collapsing into one indistinguishable object.

CONS is the computational realization of this Tangent Lambda:

$$\boxed{CONS = \Lambda_{\text{computational}}}$$

This means CONS preserves:

- CAR identity;
- CDR identity;
- ordering;
- contribution ownership;
- the selected resolution profile;
- the resulting shared projection.

The blocks meet, but their origins remain recoverable when the selected composition law permits recovery.

Tangency therefore explains why contributors may cooperate without surrendering autonomy.

7. Sparse projection instead of global materialization

The complete relation space may be extremely large.

For 32-bit CAR and CDR coordinates:

$$|CAR| = 2^{32}$$

$$|CDR| = 2^{32}$$

The complete pair space contains:

$$2^{32} \times 2^{32} = 2^{64}$$

possible ordered relations.

The system does not need to allocate a 2^{64} -cell board.

Instead, each contributor supplies a bounded projected region.

The earlier OMI design explicitly states that the shared surface is sparse and computed rather than globally materialized.

Let each source provide:

$$S_i \subseteq P_B$$

Only the active contributions are resolved:

$$B_{\text{visible}} = \Gamma(S_{\text{active}})$$

This is essential to the architecture.

The blackboard is not a giant stored universe.

It is a finite visible cross-section of a larger relational space.

8. Disjoint contributions

The cleanest blackboard condition is disjoint contribution.

For two sources:

$$S_A \cap S_B = \emptyset$$

Then their shared projection may be composed through union:

$$S_{\text{CONS}} = S_A \cup S_B$$

In bitboard form:

$$A \wedge B = 0$$

$$\text{CONS} = A \vee B$$

This is the computational form of two blocks occupying distinct regions of one shared field.

The earlier design identifies typed disjoint union as the strongest default blackboard profile. It also notes that OR is reversibly decomposable only when ownership and disjointness are known.

The geometric interpretation is:

Two autonomous blocks may share one field without interfering when their projected incidences remain disjoint.

Disjointness preserves:

- contributor ownership;
- reversible separation;
- local interpretation;
- independent update;
- visible coexistence.

9. Overlap as incidence, conflict, or agreement

Not all overlap is inherently erroneous.

For two contributions:

$$O = A \cap B$$

the overlap may mean different things depending on the declared resolver.

Prohibited overlap

If contributors are required to remain disjoint, then:

$$O \neq \emptyset$$

is a conflict.

Joint selection

If the resolver seeks agreement, the overlap may be the desired result:

$$CONS = A \cap B$$

Contrast

A resolver may expose positions selected by exactly one side:

$$CONS = A \oplus B$$

Difference

A resolver may expose what one side contributes without the other:

$$CONS = A \wedge \neg B$$

Ordered resolution

A quasigroup or another reversible operation may resolve both contributions according to a declared rule:

$$CONS = P(A) \oplus Q(B) \oplus K$$

The important principle is:

Overlap has no universal meaning outside a resolver profile.

It must not be silently discarded or automatically interpreted as failure.

10. The residual field

When projected contributions cannot be cleanly composed, the unresolved portion becomes a residual witness.

Define:

$$\mathbb{R} \subseteq \mathbb{B}$$

as the residual field.

A simple conflict residual is:

$$R_{\text{conflict}} = A \cap B$$

A mismatch residual may be:

$$R_{\text{mismatch}} = A \oplus B$$

A loss residual may record information removed during a bounded projection.

The complete result may therefore be:

$$\text{Resolve}(A, B) = (B_{\text{resolved}}, B_{\text{residual}})$$

The older blackboard specification requires overlap to remain visible through a conflict board instead of being silently destroyed.

This gives a concrete computational meaning to STDERR:

STDERR = residual incidence field

STDERR is not merely a stream of human-readable error messages.

At the geometric level, it is the surface containing:

- unresolved overlap;
- ambiguity;
- projection mismatch;
- failed closure;
- ownership conflict;
- lost information;
- rejected incidence.

11. Independent knowledge sources

A blackboard system allows multiple independent contributors to post bounded structures.

In the earlier OMI model, the primary knowledge sources were represented as separate boards:

RULES
FACTS
COMBINATORS
CLOSURES

CONS composes their visible contributions while preserving pairwise disjointness where required.

Let:

$$\mathcal{K} = \{B_R, B_F, B_C, B_L\}$$

represent:

- rules;
- facts;
- combinators;
- closures.

Their visible contributions are:

$$S_R, S_F, S_C, S_L$$

If pairwise disjoint:

$$S_i \cap S_j = \emptyset \quad \text{for } i \neq j \quad =$$

then:

$$B_{CONS} = S_R \cup S_F \cup S_C \cup S_L$$

This may be represented tetrahedrally:

V1 RULES
V2 FACTS
V3 COMBINATORS
V4 CLOSURES

Centroid CONS

The tetrahedral form is not a replacement for the blackboard.

It is the geometric explanation of four autonomous contributors resolving relative to one shared center.

12. The 256-position computational field

One bounded realization of the Tangential Block Field is the byte plane:

$$P_B = \{0x00, \dots, 0xFF\}$$

It contains 256 visible positions.

This surface is partitioned into:

$$0x00 \dots 0x7F$$

as the local field, and:

$$0x80 \dots 0xFF$$

as the remote field.

The earlier model treats bit 7 as the local/remote selector.

A coordinate may be decomposed as:

$$p = (r, h, v, s, c)$$

where:

- r selects local or remote;
- h selects the major low or high region;
- v selects one of four vertical positions;
- s selects left or right;
- c selects one of eight local columns.

The product is:

$$2 \times 2 \times 4 \times 2 \times 8 = 256$$

Thus the byte plane is not an arbitrary flat list.

It is a finite product field with eight 32-position charts.

13. Local and remote as antipodal projections

The local and remote halves may be related through:

$$a(x) = x \oplus 0x80$$

This operation is involutive:

$$a(a(x)) = x$$

Therefore local and remote may be interpreted as antipodal orientations of one shared field.

They are not required to be two unrelated blackboards.

They are two views of the same bounded coordinate structure:

$$\boxed{\mathbb{B}_{remote} = a(\mathbb{B}_{local})}$$

This preserves the principle of one centroid with opposing orientations.

14. Blackboard, graph, and canvas

The blackboard field must be distinguished from a graph and from its visual rendering.

A graph represents structure through nodes and pairwise edges.

A block design may represent higher-order incidence that is not naturally reducible to one edge at a time.

A canvas renders selected graph or block information spatially.

The older OMI canvas specification distinguishes the canonical graph from OMI-Lisp, ASCII, JSON Canvas, and graphical host projections.

The geometry-first hierarchy is:

Incidence $\rightarrow$ Block Design $\rightarrow$ Tangential Resolution $\rightarrow$ Blackboard Field $\rightarrow$ Graph Projection $\rightarrow$ Canvas Rendering

The blackboard is where contributions become jointly visible.

The graph is one structural interpretation of those visible incidences.

The canvas is one display projection of the graph or field.

Rendering must not be confused with relation resolution or acceptance.

15. Computational definition

A computational Tangential Block Field may be defined as:

$$\mathfrak{B} = (P_B, \mathcal{K}, \Pi, \Lambda, \Gamma, \mathbb{R})$$

where:

- P_B is the bounded shared coordinate surface;
- $\mathcal{K}$ is the family of autonomous knowledge blocks;
- Π projects each block into the field;
- Λ resolves tangency between selected blocks;
- Γ composes projected contributions;
- $\mathbb{R}$ records unresolved incidence.

For each knowledge source K_i :

$$S_i = \Pi(K_i)$$

For selected sources K_i, K_j :

$$T_{ij} = \Lambda(K_i, K_j)$$

The visible result is:

$$B_{\text{visible}} = \Gamma(\Pi(T_{ij}), S_1, \dots, S_n)$$

The residual is:

$$R = \text{Residual}(S_1, \dots, S_n)$$

16. Core laws

Law 1 — Contributor autonomy

A knowledge block exists independently of its blackboard projection.

$$K_i / \Pi(K_i)$$

Law 2 — Projection boundedness

Every visible contribution occupies a declared bounded region.

$$\Pi(K_i) \subseteq P_B$$

Law 3 — Identity preservation

Projection must not be treated as the complete identity unless the mapping is proven injective.

Law 4 — Declared composition

Contributions may be joined only through an explicitly selected composition profile.

Law 5 — Tangent preservation

CONS must preserve the identities and ordering of its participating components when the profile claims reversibility.

Law 6 — Residual visibility

Conflict, ambiguity, overlap, or projection loss must be exposed as a residual witness.

Law 7 — Sparse realization

The runtime need not materialize the complete Cartesian relation space.

Law 8 — View separation

The computational field, graph projection, and visual rendering are different layers.

Law 9 — Orientation relativity

Local and remote positions are interpreted relative to one shared centroid or reference field.

Law 10 — Deterministic resolution

The same active blocks, coordinates, orientations, and resolver profile must yield the same field result.

17. Canonical statement

The complete doctrine may be stated as follows:

A knowledge source is an autonomous relational block.

A blackboard contribution is a bounded projection of that block into a shared coordinate field.

CONS is the computational Tangent Lambda that resolves ordered relations among selected blocks.

The visible blackboard is the composition of active projected contributions.

The visible coordinate is a view, not the complete identity.

Disjoint contributions may coexist through typed union.

Overlapping contributions require an explicit intersection, contrast, difference, priority, or reversible resolution profile.

Unresolved incidence remains visible in the residual field.

A graph is one structural projection of the field.

A canvas is one visual rendering of that projection.

18. Final equivalence

The conventional Blackboard Pattern says:

Independent knowledge sources post partial results to a shared workspace.

Tangential Block Geometry states the same architecture more fundamentally:

Autonomous block designs project bounded incidences into a shared tangential field.

CONS resolves where and how the projected blocks meet.

The field displays their compatible relations
and preserves their unresolved residuals.

Therefore:

Knowledge Source = Autonomous Block Design

Posted Result = Bounded Field Projection

Blackboard = Shared Tangential Projection Field

CONS = Computational Tangent Lambda

Conflict Board = Residual Incidence Field

and finally:

Blackboard Pattern = Tangential Block Field viewed computationally